

ON SOME FORMULAE OF CODAZZI AND WEINGARTEN (continued)

[931, pp. 254–264]

and he writes also I for the inclination of the principal normal to the normal of the surface, $\frac{dm}{dt} dt$ for the angle of contingence, or inclination of the tangent at the point (t, τ) to the tangent at the point $(t + dt, \tau)$, and $\frac{dn}{dt} dt$ for the angle of torsion, or inclination of the osculating plane at the point (t, τ) to that at the point $(t + dt, \tau)$, or, what is the same thing, the inclination of the binormals at these points respectively. And he gives as known formulæ

$$\frac{da}{dt} = b \frac{dm}{dt}, \quad \frac{db}{dt} = -a \frac{dm}{dt} - c \frac{dn}{dt}, \quad \frac{dc}{dt} = b \frac{dn}{dt},$$

where a, b, c denote

$$(a_x, b_x, c_x), (a_y, b_y, c_y), \text{ or } (a_z, b_z, c_z).$$

He uses the capital letters

$$A_x, A_y, A_z, B_x, B_y, B_z, C_x, C_y, C_z, L, \frac{dM}{dT} dT, \frac{dN}{dT} dT,$$

with the like significations in regard to the curve for which T is variable, or say the T -curve; for greater clearness I give the accompanying figure.

[Figure: schematic showing the surface with two curves through a common point P ; tangent vectors T (along t -curve) and T' (along T -curve) lie in the tangent plane, with the inclination angle θ between them; auxiliary points B, A marked.]

Codazzi writes further

$$\begin{aligned} \frac{dm}{dt} \cos I &= u, & \frac{dm}{dt} \sin I &= v, & \frac{dn}{dt} - \frac{dI}{dt} &= w, \\ \frac{dM}{dT} \cos L &= U, & \frac{dM}{dT} \sin L &= V, & \frac{dN}{dT} - \frac{dL}{dT} &= W; \end{aligned}$$

also, if s, S are the arcs of the two curves respectively,

$$\frac{ds}{dt} = r, \quad \frac{dS}{dT} = R;$$

and he obtains a system of six formulæ, which in the *Addition*, p. 44, are presented in the following form—only I use therein θ , instead of his b , to denote the inclination of the two curves to each other:

$$\begin{aligned} \frac{du}{dT} &= \frac{dU}{dT} \cos \theta + \frac{dW}{dT} \sin \theta + w \left(V - r \frac{d\theta}{dT} \right) + (U \sin \theta - W \cos \theta) \left(v - r \frac{d\theta}{dT} \right), \\ \frac{dU}{dT} &= \frac{du}{dT} \cos \theta + \frac{dw}{dT} \sin \theta + W \left(v - r \frac{d\theta}{dT} \right) + (u \sin \theta - w \cos \theta) \left(V - R \frac{d\theta}{dT} \right), \\ \left(\frac{dv}{dT} + \frac{dV}{dT} \right) \frac{d^2 \theta}{dt dT} &+ \sin \theta (uU - wW) - \cos \theta (uW + wU) = 0, \\ R(u \cos \theta + w \sin \theta) &= r(U \cos \theta + W \sin \theta), \\ R \sin \theta \left(v - r \frac{d\theta}{dT} \right) + \frac{dR}{dT} \cos \theta &= \frac{dr}{dT}, \\ r \sin \theta \left(V - R \frac{d\theta}{dT} \right) + \frac{dr}{dT} \cos \theta &= \frac{dR}{dT}. \end{aligned}$$

In the Gaussian notation, taking p, q for the parameters, we have, with the slight variations presently referred to,

$$dx = a dp + a' dq + \frac{1}{2} \alpha dp^2 + \alpha' dp dq + \alpha'' dq^2,$$

$$\begin{aligned} dy &= b dp + b' dq + \frac{1}{2}\beta dp^2 + \beta' dp dq + \beta'' dq^2, \\ dz &= c dp + c' dq + \frac{1}{2}\gamma dp^2 + \gamma' dp dq + \gamma'' dq^2, \\ A, B, C &= bc' - b'c, ca' - c'a, ab' - a'b, \\ E, F, G &= a^2 + b^2 + c^2, aa' + bb' + cc', a'^2 + b'^2 + c'^2; \end{aligned}$$

and therefore

$$\begin{aligned} dx^2 + dy^2 + dz^2 &= E dp^2 + 2F dp dq + G dq^2; \\ E', F', G' &= A\alpha + B\beta + C\gamma, A\alpha' + B\beta' + C\gamma', A\alpha'' + B\beta'' + C\gamma''; \end{aligned}$$

and I take further

$$\begin{aligned} \omega, \omega', \omega'' &= a\alpha + b\beta + c\gamma, a\alpha' + b\beta' + c\gamma', a\alpha'' + b\beta'' + c\gamma'', \\ \varpi, \varpi', \varpi'' &= a'\alpha + b'\beta + c'\gamma, a'\alpha' + b'\beta' + c'\gamma', a'\alpha'' + b'\beta'' + c'\gamma'', \\ \lambda, \lambda', \lambda'' &= \alpha^2 + \beta^2 + \gamma^2, \alpha'^2 + \beta'^2 + \gamma'^2, \alpha''^2 + \beta''^2 + \gamma''^2, \\ \mu, \mu', \mu'' &= \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'', \alpha''\alpha + \beta''\beta + \gamma''\gamma, \alpha\alpha' + \beta\beta' + \gamma\gamma', \\ E\lambda - \omega^2 &= \Delta, \quad G\lambda'' - \varpi''^2 = \Delta'', \end{aligned}$$

where it is to be noticed that V, E', F', G' , are written instead of Gauss's Δ, D, D', D'' , and $\omega, \omega', \omega'', \varpi, \varpi', \varpi''$ instead of his m, m', m'', n, n', n'' : and that he gives for the last-mentioned quantities the values

$$\begin{aligned} m &= \frac{1}{2}\frac{dE}{dp}, \quad m' = \frac{1}{2}\frac{dE}{dq}, \quad m'' = \frac{dF}{dq} - \frac{1}{2}\frac{dG}{dp}, \\ n &= \frac{dF}{dp} - \frac{1}{2}\frac{dE}{dq}, \quad n' = \frac{1}{2}\frac{dG}{dp}, \quad n'' = \frac{1}{2}\frac{dG}{dq}, \end{aligned}$$

or, say

$$\begin{aligned} \omega &= \frac{1}{2}E_1, \quad \omega' = \frac{1}{2}E_2, \quad \omega'' = F_2 - \frac{1}{2}G_1, \\ \varpi &= F_1 - \frac{1}{2}E_2, \quad \varpi' = \frac{1}{2}G_1, \quad \varpi'' = \frac{1}{2}G_2, \end{aligned}$$

where the subscripts (1) and (2) denote differentiation in regard to p and q respectively.

Observing that the cosine-inclinations of the tangent to the p -curve are as a, b, c , and those of the binormal or perpendicular to the osculating plane are as $b\gamma - c\beta, c\alpha - a\gamma, a\beta - b\alpha$, we easily find

$$\begin{aligned} a_x, a_y, a_z &= \frac{a}{\sqrt{E}}, \frac{b}{\sqrt{E}}, \frac{c}{\sqrt{E}}, \\ c_x, c_y, c_z &= \frac{b\gamma - c\beta}{\sqrt{\Delta}}, \frac{c\alpha - a\gamma}{\sqrt{\Delta}}, \frac{a\beta - b\alpha}{\sqrt{\Delta}}, \\ b_x, b_y, b_z &= \frac{E\alpha - a\omega}{\sqrt{E\Delta}}, \frac{E\beta - b\omega}{\sqrt{E\Delta}}, \frac{E\gamma - c\omega}{\sqrt{E\Delta}}, \end{aligned}$$

and for the cosine-inclinations of the normal of the surface

$$\Delta_x, \Delta_y, \Delta_z = \frac{A}{V}, \frac{B}{V}, \frac{C}{V},$$

we find

$$\begin{aligned} \cos I &= \frac{A(b\gamma - c\beta) + B(c\alpha - a\gamma) + C(a\beta - b\alpha)}{V\sqrt{\Delta}} \\ &= \frac{(a^2 + b^2 + c^2)(a'\alpha + b'\beta + c'\gamma) - (aa' + bb' + cc')(a\alpha + b\beta + c\gamma)}{V\sqrt{\Delta}} \\ &= \frac{E\varpi - F\omega}{V\sqrt{\Delta}}, \\ \sin I &= \frac{A(E\alpha - a\omega) + B(E\beta - b\omega) + C(E\gamma - c\omega)}{V\sqrt{E\Delta}}; \end{aligned}$$

or, since $Aa + Bb + Cc = 0$, this is

$$\sin I = \frac{E(A\alpha + B\beta + C\gamma)}{V\sqrt{E\Delta}} = \frac{E'\sqrt{E}}{V\sqrt{\Delta}}.$$

We ought therefore to have

$$E'^2 E + (E\varpi - F\omega)^2 = V^2 \Delta = (EG - F^2)(E\lambda - \omega^2),$$

or, omitting the terms $F^2\omega^2$, which destroy each other, and throwing out a factor E ,

$$E\varpi^2 - 2F\varpi\omega + G\omega^2 - \lambda(EG - F^2) = \lambda(EG - F^2) - E'^2;$$

viz., putting this for $EG - F^2$ its value, $= A^2 + B^2 + C^2$, and for the other terms their values, this is

$$\begin{aligned} & (a^2 + b^2 + c^2)(a'\alpha + b'\beta + c'\gamma)^2 - 2(aa' + bb' + cc')(a'\alpha + b'\beta + c'\gamma)(a\alpha + b\beta + c\gamma) + (a'^2 + b'^2 + c'^2)(a\alpha + b\beta + c\gamma)^2 \\ & = (A^2 + B^2 + C^2)(\alpha^2 + \beta^2 + \gamma^2) - (A\alpha + B\beta + C\gamma)^2. \end{aligned}$$

The right-hand side is here $= (B\gamma - C\beta)^2 + (C\alpha - A\gamma)^2 + (A\beta - B\alpha)^2$: the left-hand consists of three parts, the first whereof is

$$\{a(a'\alpha + b'\beta + c'\gamma) - a'(a\alpha + b\beta + c\gamma)\}^2 = (-B\gamma + C\beta)^2,$$

and similarly the other two parts are

$$(-C\alpha + A\gamma)^2 \text{ and } (-A\beta + B\alpha)^2;$$

the equation is thus verified.

We require Codazzi's $\frac{dm}{dt}$ and $\frac{dn}{dt}$, or say $\frac{dm}{dp}$ and $\frac{dn}{dp}$; these are to be obtained from the equations

$$\frac{d}{dp} \frac{a}{\sqrt{E}} = \frac{E\alpha - a\omega}{\sqrt{E\Delta}} \frac{dm}{dp}, \quad \frac{d}{dp} \frac{b\gamma - c\beta}{\sqrt{\Delta}} = \frac{E\alpha - a\omega}{\sqrt{E\Delta}} \frac{dn}{dp},$$

and the values obtained should satisfy

$$\frac{d}{dp} \frac{E\alpha - a\omega}{\sqrt{E\Delta}} = -\frac{a}{\sqrt{E}} \frac{dm}{dp} - \frac{b\gamma - c\beta}{\sqrt{\Delta}} \frac{dn}{dp}.$$

I find

$$\frac{dm}{dp} = \frac{\sqrt{\Delta}}{E}, \quad \frac{dn}{dp} = \frac{\sqrt{E}}{\Delta} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix},$$

where $\alpha_1, \beta_1, \gamma_1$ are the derivatives of α, β, γ in regard to p ; and the equation to be verified thus is

$$\frac{d}{dp} \frac{E\alpha - a\omega}{\sqrt{E\Delta}} = -\frac{a\sqrt{\Delta}}{E\sqrt{E}} + \frac{(b\gamma - c\beta)\sqrt{E}}{\Delta\sqrt{\Delta}} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}.$$

First, for $\frac{dm}{dp}$: the derivatives of a, E are α and $2(a\alpha + b\beta + c\gamma) = 2\omega$; we thus have

$$\frac{d}{dp} \frac{a}{\sqrt{E}} = \frac{\alpha}{\sqrt{E}} - \frac{a\omega}{E\sqrt{E}} = \frac{E\alpha - a\omega}{E\sqrt{E}},$$

which is

$$= \frac{E\alpha - a\omega}{\sqrt{E\Delta}} \frac{dm}{dp},$$

viz. we have

$$\frac{dm}{dp} = \frac{\sqrt{\Delta}}{\sqrt{E\sqrt{E}}} \quad \text{or} \quad \frac{dm}{dp} = \frac{\sqrt{\Delta}}{E}.$$

Next, for $\frac{dn}{dp}$: using a subscript (1) to denote derivation in regard to p , we have $\Delta = E\lambda - \omega^2$, and thence

$$\begin{aligned}\Delta_1 &= E_1\lambda + E\lambda_1 - 2\omega\omega_1, \\ &= 2a\alpha \cdot \lambda + 2E(\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1) - 2\omega(\lambda + a\alpha_1 + b\beta_1 + c\gamma_1), \\ &= 2\{E(\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1) - \omega(a\alpha_1 + b\beta_1 + c\gamma_1)\} \\ &= 2(\alpha_1X + \beta_1Y + \gamma_1Z),\end{aligned}$$

if, for a moment,

$$X, Y, Z = E\alpha - a\omega, \quad E\beta - b\omega, \quad E\gamma - c\omega;$$

these values give identically

$$\alpha X + \beta Y + \gamma Z = \Delta, \quad \text{and} \quad aX + bY + cZ = 0.$$

Hence we have

$$\frac{d}{dp} \frac{b\gamma - c\beta}{\sqrt{\Delta}} = \frac{b\gamma_1 - c\beta_1}{\sqrt{\Delta}} - \frac{(b\gamma - c\beta)(\alpha_1X + \beta_1Y + \gamma_1Z)}{\Delta\sqrt{\Delta}},$$

which must be

$$= \frac{X}{\sqrt{E\Delta}} \frac{dn}{dp};$$

we thus have

$$(b\gamma_1 - c\beta_1)\Delta - (b\gamma - c\beta)(\alpha_1X + \beta_1Y + \gamma_1Z) = \frac{\Delta X}{\sqrt{E}} \frac{dn}{dp},$$

or, putting the left-hand side in the form

$$(b\gamma_1 - c\beta_1)(\alpha X + \beta Y + \gamma Z) - (b\gamma - c\beta)(\alpha_1X + \beta_1Y + \gamma_1Z) - (\beta\gamma_1 - \beta_1\gamma)(aX + bY + cZ),$$

this is

$$= -X \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix},$$

and we thus find

$$\frac{dn}{dp} = \frac{\sqrt{E}}{\Delta} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}.$$

For the verification of the equation

$$\frac{d}{dp} \frac{E\alpha - a\omega}{\sqrt{E\Delta}} = -\frac{a\sqrt{\Delta}}{E\sqrt{E}} + \frac{(b\gamma - c\beta)\sqrt{E}}{\Delta\sqrt{\Delta}} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix},$$

we have

$$(E\alpha - a\omega)_1 = E_1\alpha - a(a\alpha + b\beta + c\gamma)_1 + a\omega - a\lambda,$$

$$(E\Delta)_1 = 2E\omega(\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1) - 2E\omega(a\alpha_1 + b\beta_1 + c\gamma_1) + 2\Delta\omega,$$

and hence the equation $(E\Delta)^{3/2} \frac{d}{dp} \frac{E\alpha - a\omega}{\sqrt{E\Delta}}$ becomes

$$\begin{aligned}&\{E(\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1) - \omega(a\alpha_1 + b\beta_1 + c\gamma_1) + \omega\}(E\alpha - a\omega) + E\lambda - a(a\alpha_1 + b\beta_1 + c\gamma_1) + \omega \\ &= \frac{(b\gamma - c\beta)\sqrt{E}}{\sqrt{\Delta}} \cdot \sqrt{E}\sqrt{\Delta} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}.\end{aligned}$$

Considering first the terms without $\alpha_1, \beta_1, \gamma_1$, these give

$$-\omega(E\alpha - a\omega) + E(a\omega - a\lambda) = -a(E\lambda - \omega^2),$$

which is identically true; and then the remaining terms with $\alpha_1, \beta_1, \gamma_1$ give

$$\begin{aligned} & -\{E(\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1) - \omega(a\alpha_1 + b\beta_1 + c\gamma_1)\}(E\alpha - a\omega) + \{E\lambda - a(a\alpha_1 + b\beta_1 + c\gamma_1)\}(E\lambda - \omega^2) \\ & = (b\gamma - c\beta)E \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}. \end{aligned}$$

On the left-hand side, the whole coefficient of α_1 is

$$= -(E\alpha - a\omega)a + (b^2 + c^2)(E\lambda - \omega^2),$$

which is

$$\begin{aligned} & = -E\alpha^2 + 2a\alpha\omega - a^2\omega^2 + (b^2 + c^2)E\lambda - (b^2 + c^2)\omega^2, \\ & = E[-E\alpha^2 + 2a\alpha\omega + (b^2 + c^2)\lambda - \omega^2]; \end{aligned}$$

and, substituting for E, ω , and λ their values, this is found to be

$$= E\{(b^2 + c^2)(\beta^2 + \gamma^2) - (b\beta + c\gamma)^2\} = E(b\gamma - c\beta)^2.$$

Similarly the whole coefficients of β_1 and γ_1 are found to be

$$= E(b\gamma - c\beta)(c\alpha - a\gamma), \quad \text{and} \quad E(b\gamma - c\beta)(a\beta - b\alpha),$$

respectively; and thus the left-hand side becomes

$$= (b\gamma - c\beta)E \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix},$$

as it should do.

We have

$$\tan I = \frac{P}{Q},$$

where

$$\begin{aligned} P &= E'\sqrt{E} = \sqrt{E}(A\alpha + B\beta + C\gamma), & &= a\alpha + b\beta + c\gamma, \text{ suppose,} \\ Q &= E\varpi - F\omega = (cB - bC)\alpha + (aC - cA)\beta + (bA - aB)\gamma, & &= a'\alpha + b'\beta + c'\gamma, \text{ suppose,} \\ & & & P^2 + Q^2 = V^2\Delta, \end{aligned}$$

and we have hence to find $\frac{dI}{dp}$. Using, as before, a subscript (1) to denote derivation in regard to p , we have

$$\frac{dI}{dp} = \frac{QP_1 - PQ_1}{P^2 + Q^2};$$

the numerator is

$$\begin{aligned} & = (a'\alpha + b'\beta + c'\gamma)(a\alpha_1 + b\beta_1 + c\gamma_1 + a_1\alpha + b_1\beta + c_1\gamma) - (a\alpha + b\beta + c\gamma)(a'\alpha_1 + b'\beta_1 + c'\gamma_1 + a'_1\alpha + b'_1\beta + c'_1\gamma) \\ & = -[(bc' - b'c)(\beta\gamma_1 - \beta_1\gamma) + (ca' - c'a)(\gamma\alpha_1 - \gamma_1\alpha) + (ab' - a'b)(\alpha\beta_1 - \alpha_1\beta)] + Q(a_1\alpha + b_1\beta + c_1\gamma) - P(a'_1\alpha + b'_1\beta + c'_1\gamma). \end{aligned}$$

For the first part hereof,

$$\begin{aligned} bc' - b'c &= \sqrt{E}\{B(bA - aB) - C(aC - cA)\} \\ &= \sqrt{E}\{A(aA + bB + cC) - a(A^2 + B^2 + C^2)\} = -aV^2\sqrt{E}, \end{aligned}$$

since $aA + bB + cC = 0$; and similarly

$$ca' - c'a, \quad ab' - a'b = -bV^2\sqrt{E}, \quad -cV^2\sqrt{E},$$

respectively; and thus the first part is

$$\begin{aligned} &= V^2 \sqrt{E} \{a(\beta\gamma_1 - \beta_1\gamma) + b(\gamma\alpha_1 - \gamma_1\alpha) + c(\alpha\beta_1 - \alpha_1\beta)\}, \\ &= V^2 \sqrt{E} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}. \end{aligned}$$

Hence, dividing by

$$P^2 + Q^2 = V^2 \Delta,$$

the first part of $\frac{dI}{dp}$ is

$$= \frac{\sqrt{E}}{\Delta} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix}.$$

For the second part of the numerator, we require

$$\alpha\alpha_1 + \beta\beta_1 + \gamma\gamma_1 \quad \text{and} \quad \alpha\alpha'_1 + \beta\beta'_1 + \gamma\gamma'_1;$$

the values of a, b, c are $A\sqrt{E}, B\sqrt{E}, C\sqrt{E}$, where $E_1 = 2\omega$, and hence

$$a_1\alpha + b_1\beta + c_1\gamma = (A_1\alpha + B_1\beta + C_1\gamma)\sqrt{E} + \frac{\omega}{\sqrt{E}}(A\alpha + B\beta + C\gamma).$$

From the values

$$A, B, C = bc' - b'c, ca' - c'a, ab' - a'b,$$

we have

$$A_1, B_1, C_1 = \beta c' - \gamma b' + b\gamma' - c\beta', \gamma a' - \alpha c' + c\alpha' - a\gamma', \alpha b' - \beta a' + a\beta' - b\alpha',$$

and thence

$$\begin{aligned} A_1\alpha + B_1\beta + C_1\gamma &= -[a(\beta\gamma' - \beta'\gamma) + b(\gamma\alpha' - \gamma'\alpha) + c(\alpha\beta' - \alpha'\beta)], \\ &= - \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix}. \end{aligned}$$

Hence

$$a_1\alpha + b_1\beta + c_1\gamma = -\sqrt{E} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} + \frac{\omega E'}{\sqrt{E}}.$$

Next, we have

$$a', b', c' = cB - bC, aC - cA, bA - aB,$$

and thence

$$\begin{aligned} \alpha\alpha'_1 + \beta\beta'_1 + \gamma\gamma'_1 &= \alpha(\gamma B - \beta C + bC_1 - cB_1) + \beta(\alpha C - \gamma A + cA_1 - aC_1) + \gamma(\beta A - \alpha B + aB_1 - bA_1) \\ &= (b\gamma - c\beta)A_1 + (c\alpha - a\gamma)B_1 + (a\beta - b\alpha)C_1 \\ &= (b\gamma - c\beta)(c'\beta - b'\gamma + b\gamma' - c\beta') \\ &\quad + (c\alpha - a\gamma)(a'\gamma - c'\alpha + c\alpha' - a\gamma') \\ &\quad + (a\beta - b\alpha)(b'\alpha - a'\beta + a\beta' - b\alpha'); \end{aligned}$$

the portion hereof, which is quadric in α, β, γ , is

$$\begin{aligned} &= -(\alpha^2 + \beta^2 + \gamma^2)(aa' + bb' + cc') + (a\alpha + b\beta + c\gamma)(a'\alpha + b'\beta + c'\gamma), \\ &= -\lambda F + \varpi\omega, \end{aligned}$$

and the remaining portion, which is lineo-linear in α, β, γ and α', β', γ' , is

$$= (\alpha\alpha' + \beta\beta' + \gamma\gamma')(a^2 + b^2 + c^2) - (a\alpha + b\beta + c\gamma)(a\alpha' + b\beta' + c\gamma'),$$

$$= E\mu'' - \omega\omega';$$

we thus have

$$\alpha a'_1 + \beta b'_1 + \gamma c'_1 = -\lambda F + E\mu'' + \omega(\varpi - \omega').$$

Hence the second portion of the numerator, or

$$Q(\alpha a_1 + \beta b_1 + \gamma c_1) - P(\alpha a'_1 + \beta b'_1 + \gamma c'_1),$$

is

$$= (E\varpi - F\omega) \left\{ -\sqrt{E} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} + \frac{\omega E'}{\sqrt{E}} \right\} - E'\sqrt{E} \{ \mu'' E - \lambda F + \omega(\varpi - \omega') \}.$$

There are two terms, $+E'\sqrt{E}\omega\varpi$ and $-E'\sqrt{E}\omega\varpi$, which destroy each other, and thus the whole second part of the numerator is

$$= -(E\varpi - F\omega) \sqrt{E} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} - E'\sqrt{E}(\mu'' E - \lambda F) + \frac{E'\omega}{\sqrt{E}}(\omega' E - \omega F),$$

and, for the corresponding part of $\frac{dI}{dp}$, we must divide this by $P^2 + Q^2 = V^2\Delta$.

Hence, finally, we have

$$\frac{dI}{dp} = \frac{\sqrt{E}}{\Delta} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix} + \frac{1}{V^2\Delta} \left\{ -(E\varpi - F\omega) \sqrt{E} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} - E'\sqrt{E}(\mu'' E - \lambda F) + \frac{E'\omega}{\sqrt{E}}(\omega' E - \omega F) \right\}.$$

where I recall that the values of ω , ϖ , ω' , λ , and μ'' are $a\alpha + b\beta + c\gamma$, $a'\alpha + b'\beta + c'\gamma$, $a\alpha' + b\beta' + c\gamma'$, $\alpha^2 + \beta^2 + \gamma^2$, and $\alpha\alpha' + \beta\beta' + \gamma\gamma'$, respectively. Observe that the first term in this expression for $\frac{dI}{dp}$ is

$$= -\frac{dn}{dp}.$$

We thus have

$$u = \frac{dm}{dp} \cos I = \frac{E\varpi - F\omega}{V\sqrt{E}},$$

$$v = \frac{dm}{dp} \sin I = \frac{E'}{V\sqrt{E}},$$

$$w = \frac{dn}{dp} - \frac{dI}{dp} = \frac{2\sqrt{E}}{\sqrt{\Delta}} \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha_1, & \beta_1, & \gamma_1 \end{vmatrix} + \frac{1}{V^2\Delta} \left\{ (\varpi E - \omega F) \begin{vmatrix} a, & b, & c \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} \sqrt{E} + E'\sqrt{E}(\mu'' E - \lambda F) - \frac{E'\omega}{\sqrt{E}}(\omega' E - \omega F) \right\},$$

and we thence obtain at once the values of U , V , W ; viz. these are

$$U = \frac{dM}{dq} \cos L = \frac{G\varpi'' - F\lambda''}{V\sqrt{G}},$$

$$V = \frac{dM}{dq} \sin L = \frac{G'}{V\sqrt{G}},$$

$$W = \frac{dN}{dq} - \frac{dL}{dq} = \frac{2\sqrt{G}}{\sqrt{\Delta''}} \begin{vmatrix} a', & b', & c' \\ \alpha'', & \beta'', & \gamma'' \\ \alpha'_2, & \beta'_2, & \gamma'_2 \end{vmatrix} + \frac{1}{V^2\Delta''} \left\{ (\varpi'' G - \omega'' F) \begin{vmatrix} a', & b', & c' \\ \alpha'', & \beta'', & \gamma'' \\ \alpha', & \beta', & \gamma' \end{vmatrix} \sqrt{G} + \dots \right\},$$

where α''_2 , β''_2 , γ''_2 denote the derived functions of α'' , β'' , γ'' in regard to q , viz. these are the third derived functions of x , y , z in regard to q .

We have, moreover,

$$dx^2 + dy^2 + dz^2 = E dp^2 + 2F dp dq + G dq^2 = r^2 dt^2 + 2rR \cos \theta dt dT + R^2 dT^2;$$

that is,

$$r = \sqrt{E}, \quad R = \sqrt{G}, \quad \cos \theta = \frac{F}{\sqrt{EG}},$$

and therefore also

$$\sin \theta = \frac{\sqrt{EG - F^2}}{\sqrt{EG}} = \frac{V}{\sqrt{EG}}.$$

Writing p, q in place of t, T , and for r, R substituting their values, then, with the foregoing values of u, v, w, U, V, W , Codazzi's six equations are

$$\begin{aligned} \frac{du}{dq} &= \frac{dU}{dp} \cos \theta + \frac{dW}{dp} \sin \theta + w \left(V - \sqrt{E} \frac{d\theta}{dq} \right) + (U \sin \theta - W \cos \theta) \left(v - \sqrt{E} \frac{d\theta}{dp} \right), \\ \frac{dU}{dp} &= \frac{du}{dq} \cos \theta + \frac{dw}{dq} \sin \theta + W \left(v - \sqrt{E} \frac{d\theta}{dp} \right) + (u \sin \theta - w \cos \theta) \left(V - \sqrt{G} \frac{d\theta}{dq} \right), \\ \left(\frac{dv}{dq} + \frac{dV}{dp} \right) \frac{d^2 \theta}{dp dq} &+ \sin \theta (uU - wW) - \cos \theta (uW + wU) = 0, \\ \sqrt{G} (u \cos \theta + w \sin \theta) &= \sqrt{E} (U \cos \theta + W \sin \theta), \\ \sqrt{G} \sin \theta \left(v - \sqrt{E} \frac{d\theta}{dp} \right) &+ \frac{d\sqrt{G}}{dp} \cos \theta = \frac{d\sqrt{E}}{dq}, \\ \sqrt{E} \sin \theta \left(V - \sqrt{G} \frac{d\theta}{dq} \right) &+ \frac{d\sqrt{E}}{dq} \cos \theta = \frac{d\sqrt{G}}{dp}. \end{aligned}$$

I take the opportunity of remarking that Gauss, in §11 of his memoir, gives the first and second of the formulas

$$\begin{aligned} \alpha V^2 + a(\varpi F - \omega G) + a'(\omega F - \varpi E) - AE' &= 0, \\ \alpha' V^2 + a(\varpi' F - \omega' G) + a'(\omega' F - \varpi' E) - AF' &= 0, \\ \alpha'' V^2 + a(\varpi'' F - \omega'' G) + a'(\omega'' F - \varpi'' E) - AG' &= 0, \end{aligned}$$

(each of them one out of a system of three like equations), where, as before,

$$\begin{aligned} E', F', G' &= A\alpha + B\beta + C\gamma, \quad A\alpha' + B\beta' + C\gamma', \quad A\alpha'' + B\beta'' + C\gamma'', \\ \omega, \omega', \omega'' &= a\alpha + b\beta + c\gamma, \quad a\alpha' + b\beta' + c\gamma', \quad a\alpha'' + b\beta'' + c\gamma'', \\ \varpi, \varpi', \varpi'' &= a'\alpha + b'\beta + c'\gamma, \quad a'\alpha' + b'\beta' + c'\gamma', \quad a'\alpha'' + b'\beta'' + c'\gamma''. \end{aligned}$$

These are, in fact, the formulas (IV.),

$$\begin{aligned} \frac{d^2 x}{dp^2} &= \binom{11}{1} \frac{dx}{dp} + \binom{11}{2} \frac{dx}{dq} + c_{11} X = 0, \\ \frac{d^2 x}{dp dq} &= \binom{12}{1} \frac{dx}{dp} + \binom{12}{2} \frac{dx}{dq} + c_{12} X = 0, \\ \frac{d^2 x}{dq^2} &= \binom{22}{1} \frac{dx}{dp} + \binom{22}{2} \frac{dx}{dq} + c_{22} X = 0, \end{aligned}$$

of the memoir, Weingarten, “*Ueber die Deformation einer biegsamen unausdehnbaren Fläche*,” *Crelle*, t. c. (1887), pp. 296–310; viz. the symbols correspond to those of Weingarten, as follows

$$\begin{aligned} E, F, G &= a_{11}, a_{12}, a_{22}, \\ E', F', G' &= -c_{11}\sqrt{a}, -c_{12}\sqrt{a}, -c_{22}\sqrt{a}, \\ A, B, C &= PX, PY, PZ, \\ \omega, \omega', \omega'' &= \frac{1}{2} \frac{da_{11}}{dp}, \quad \frac{1}{2} \frac{da_{11}}{dq}, \quad \frac{da_{12}}{dq} - \frac{1}{2} \frac{da_{22}}{dp}, \end{aligned}$$

$$\varpi, \varpi', \varpi'' = \frac{da_{12}}{dp} - \frac{1}{2} \frac{da_{11}}{dq}, \quad \frac{1}{2} \frac{da_{22}}{dp}, \quad \frac{1}{2} \frac{da_{22}}{dq},$$

and thus Weingarten's symbols, $\binom{11}{1}$, &c., have the values

$$\binom{11}{1} = \frac{1}{a} \left[a_{22} \left(-\frac{1}{2} \frac{da_{11}}{dp} + \frac{da_{12}}{dq} \right) + a_{12} \left(\frac{1}{2} \frac{da_{11}}{dp} \right) \right],$$

$$\binom{12}{1} = \frac{1}{a} \left[a_{22} \left(-\frac{1}{2} \frac{da_{22}}{dp} \right) + a_{12} \left(\frac{1}{2} \frac{da_{22}}{dq} \right) \right],$$

$$\binom{22}{1} = \frac{1}{a} \left[a_{11} \left(-\frac{1}{2} \frac{da_{22}}{dp} + \frac{da_{12}}{dq} \right) + a_{12} \left(\frac{1}{2} \frac{da_{22}}{dp} \right) \right],$$

$$\binom{11}{2} = \frac{1}{a} \left[a_{11} \left(-\frac{1}{2} \frac{da_{11}}{dq} + \frac{da_{12}}{dp} \right) + a_{12} \left(\frac{1}{2} \frac{da_{11}}{dp} \right) \right],$$

$$\binom{12}{2} = \frac{1}{a} \left[a_{11} \left(-\frac{1}{2} \frac{da_{11}}{dq} \right) + a_{12} \left(\frac{1}{2} \frac{da_{11}}{dq} \right) \right],$$

$$\binom{22}{2} = \frac{1}{a} \left[a_{11} \left(-\frac{1}{2} \frac{da_{22}}{dq} + \frac{da_{12}}{dp} \right) + a_{12} \left(\frac{1}{2} \frac{da_{22}}{dq} \right) \right],$$

values which give, as they should do,

$$\binom{11}{1} + \binom{12}{2} = \frac{1}{\sqrt{a}} \frac{d\sqrt{a}}{dp}, \quad \binom{12}{1} + \binom{22}{2} = \frac{1}{\sqrt{a}} \frac{d\sqrt{a}}{dq}.$$

The foregoing comparison serves to explain the notation of Weingarten's valuable memoir.

ON SYMMETRIC FUNCTIONS AND SEMINVARIANTS.

[932]

From the *American Journal of Mathematics*, vol. xv. (1893), pp. 1–74.

The principal object of the present memoir is to develop further the theory of seminvariants, but in connexion therewith I was led to some investigations on symmetric functions, and I have consequently included this subject in the title. The two theories, if we adopt the MacMahon form of equation,

$$\Omega = 1 + bx + \frac{c}{2}x^2 + \frac{d}{6}x^3 + \dots,$$

may be regarded as identical; but there are still two branches of the theory, viz. we may seek to obtain for the symmetric functions of the roots expressions in terms of the coefficients (which expressions, in the case of non-unitary symmetric functions, are in fact seminvariants), or we may attend to the properties of the functions of the coefficients thus obtained and which we call seminvariants. But I do not in the first instance use the MacMahon form, but retain the ordinary form of equation $0 = 1 + bx + cx^2 + dx^3 + \dots$, and we have thus only a parallelism of the two theories, and in place of seminvariants we have functions which I call non-unitarians. In regard as well to these as to unitarian functions, I consider certain operators Θ_σ , Δ , $P - \delta b$, and $Q - 2\omega b$, which under altered forms present themselves also in the theory of seminvariants.

As regards seminvariants, I consider what I call the blunt and sharp forms respectively: the great problem is, it appears to me, that of sharp seminvariants, otherwise the *I*-and-*F* problem—viz. for any given weight we have to determine the correspondence between the initial and final terms in such wise as to obtain a system of sharp seminvariants. I obtain a “square diagram” solution, which is so far theoretically complete that for any given weight I can, without any tentative operation, determine by a laborious process the correspondence in question: but I am not thereby enabled to establish or enunciate for successive weights any general rule of correspondence; and my process is in fact, as regards practicability, far inferior to that which I call the MacMahon linkage, but of the validity of this I have not succeeded in obtaining any satisfactory proof.

I establish an umbral theory of seminvariants which will be presently again referred to, and I consider the question of the reduction of seminvariants. The final term of a seminvariant may be composite (that is, the product of two or more final terms), and that in one way only or in two or more ways, or it may be non-composite. In the case of a composite final term the seminvariant is reducible, but the converse theorem that a seminvariant with a non-composite final term is irreducible is in nowise true; the reason of this is explained. An irreducible seminvariant is a perpetuant. In regard to perpetuants, I reproduce and simplify a demonstration recently obtained by Dr Stroh as to the perpetuants for any given degree whatever: viz. the generating function for perpetuants of degree n is

$$= \frac{x^{2^n-1}}{(1-x)(1-x^2)\dots(1-x^n)};$$

the theorem was previously known, and more or less completely proved, for the values $n = 4, 5, 6$, and 7 . Dr Stroh’s investigation is conducted by an umbral representation,

$$(\alpha x + \beta y + \gamma z + \dots)^n, \quad x + y + z + \dots = 0,$$

of the blunt seminvariants of a given weight.

I consider in regard to seminvariants the theory of the symbols $P - \delta b$ and $Q - 2\omega b$, and the derived symbols Y and Z , each of which operating on a seminvariant gives a seminvariant. These are, in fact, connected with the derivatives (f, F) of a quantic f and any covariant thereof F ; but except to point out this connexion, I do not in the present memoir consider the theory of covariants.

The Coefficients $(a, b, c, d, e, \dots)$ or $(1, b, c, d, e, \dots)$. Art. Nos. 1 to 9.

1. I consider the series $(a, b, c, d, e, \dots)$, or putting as we most frequently do $a = 1$, say the series $(1, b, c, d, e, \dots)$ of coefficients, the several terms whereof are taken to be of the weights $0, 1, 2, 3, 4, \dots$ respectively. We form with these sets of isobaric terms, or say columns of the weights $0, 1, 2, 3, 4, \dots$ respectively, for instance,

0	1	2	3	4	5	6
1	b	c	d	e	f	g
		b^2	bc	bd	be	bf
			b^3	c^2	cd	ce
				b^2c	b^2d	d^2
				b^4	bc^2	b^2e
					b^3c	bcd
					b^5	c^3
						b^3d
						b^2c^2
						b^4c
						b^6

and generally a set or column of any given weight. In each term, the letters are written in alphabetical order.

Taking the whole or any part of a column, for instance the whole column (d, bc, b^3) , or the part (e, bd, c^2) of the next column, we may by supplying powers of a in such wise as to leave unaltered the terms of the highest degree, that is, by reading these as (a^2d, abc, b^3) and (ae, bd, c^2) respectively, regard them as homogeneous sets of a given degree in $(a, b, c, d, e, \dots)$; and thus generally we may speak of the degree of a set of terms.

The terms of the several columns as above written down are in alphabetical order, AO ; viz. we supply as above the proper powers of a , reading for instance col. 4 as $a^3e, a^2bd, a^2c^2, ab^2c, b^4$, where the terms are in alphabetical or dictionary order.

Each column is derived from the preceding one by Arbogast’s rule, it being understood, for instance, that b^4 , that is, ab^4 , gives the two terms ab^3c and b^5 , that is, b^3c and b^5 ; and so in other cases.

2. We attend in particular to the non-unitary terms, or non-unitaries, e.g. in col. 5, f, cd , which contain no b ; and to the power-ending terms or power-enders, bc^2, b^5 , which end in a power. It will be observed that, whenever by Arbogast’s rule a term in one column gives two terms in the next column, the second of these is a power-enders; and thus in any column the excess of the number of terms above that in the preceding column is equal to the number of power-enders.

3. I consider the notion of conjugate terms: representing, for instance, the terms

$$f, bc, cd$$

by dots in the form

$$\begin{array}{ccc} \bullet & \cdot & \cdot \\ \bullet & \bullet & \cdot \\ \bullet & \bullet & \bullet \end{array}$$

and reading the number of dots in columns instead of in lines we derive the conjugate terms

$$b^5, b^3c, bc^2,$$

and so in other cases. It is clear that the relation is a reciprocal one (thus the conjugates of b^5, b^3c, bc^2 are f, bc, cd respectively). Moreover, a term may be its own conjugate; thus cd^2 , arranging the dots in lines and reading them in columns, $\begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \end{array}$ is again cd^2 .

It is at once seen that non-unitaries and power-enders are conjugate to each other; hence in any column, the non-unitaries and the power-enders are equal in number, and a preceding result may be stated in the more complete form: in any column the excess of the number of terms above that in the preceding column is equal to the number of non-unitaries or to the number of power-enders.

4. The terms of the several columns may be arranged in counter-order CO , thus:

0	1	2	3	4	5	6
1	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
		<i>b</i> ²	<i>bc</i>	<i>bd</i>	<i>be</i>	<i>bf</i>
			<i>b</i> ³	<i>c</i> ²	<i>cd</i>	<i>ce</i>
				<i>b</i> ² <i>c</i>	<i>b</i> ² <i>d</i>	<i>d</i> ²
				<i>b</i> ⁴	<i>bc</i> ²	<i>b</i> ² <i>e</i>
					<i>b</i> ³ <i>c</i>	<i>bcd</i>
					<i>b</i> ⁵	<i>b</i> ³ <i>d</i>
						<i>c</i> ³
						<i>b</i> ² <i>c</i> ²
						<i>b</i> ⁴ <i>c</i>
						<i>b</i> ⁶

viz. we arrange here according to the highest letters. The counter-order is, in fact, the alphabetical order with the reversed arrangement ($\dots, g, f, e, d, c, b, a$) of the alphabet, but in the separate terms we retain the alphabetical order, thus writing as before bf and not fb . Observe that the difference between the two arrangements, AO and CO , first presents itself in the col. 6.

In this CO arrangement, each column is derived from the next preceding one by a rule as follows: We operate on the lowest letter of each term, being a simple letter, not a power, by changing it into the next highest letter, and we further operate upon each term by multiplying it by b , the operation or (as the case may be) two operations upon any term being performed before operating upon the next term.

5. If we compare a column in AO with the same column in CO , for instance

<i>AO</i>	<i>CO</i>	<i>AO</i>	<i>CO</i> rev.
<i>g</i>	<i>g</i>	<i>g</i>	<i>b</i> ⁶
<i>bf</i>	<i>bf</i>	<i>bf</i>	<i>b</i> ⁴ <i>c</i>
<i>ce</i>	<i>ce</i>	<i>ce</i>	<i>b</i> ² <i>c</i> ²
<i>d</i> ²	<i>d</i> ²	<i>d</i> ²	<i>c</i> ³
<i>b</i> ² <i>e</i>	<i>b</i> ² <i>e</i>	<i>b</i> ² <i>e</i>	<i>b</i> ³ <i>d</i>
<i>bcd</i>	<i>bcd</i>	<i>bcd</i>	<i>bcd</i>
<i>b</i> ³ <i>d</i>	<i>b</i> ³ <i>d</i>	<i>b</i> ³ <i>d</i>	<i>b</i> ² <i>e</i>
<i>c</i> ³	<i>c</i> ³	<i>c</i> ³	<i>d</i> ²
<i>b</i> ² <i>c</i> ²	<i>b</i> ² <i>c</i> ²	<i>b</i> ² <i>c</i> ²	<i>ce</i>
<i>b</i> ⁴ <i>c</i>	<i>b</i> ⁴ <i>c</i>	<i>b</i> ⁴ <i>c</i>	<i>bf</i>
<i>b</i> ⁶	<i>b</i> ⁶	<i>b</i> ⁶	<i>g</i>

it will be seen that the terms are conjugates of each other, the first and last, the second and last but one terms, and so on; or, what is the same thing, if we reverse the order of either column, then the pairs of conjugate terms will appear each in the same line; of course, here a self-conjugate term such as bcd is put in evidence.

6. By writing $a, b, c, d, \dots = a_0, a_1, a_2, a_3, \dots$, or more simply $0, 1, 2, 3, \dots$, we connect the theory with that of the partition of numbers: in particular, the terms of a given weight correspond to the partitions of that weight, or number of ways in which that weight can be made up with the parts $1, 2, 3, \dots$. It may be remarked that, in a partition, the parts are usually written in decreasing order, whereas (as remarked above) in a literal term the letters are written in alphabetical order. Thus we have 321 and bcd ; it would be more correct to write the partition as 123 .

It is frequently convenient, retaining the letters $b, c, d, \dots$, to write for instance $q = a_\sigma$ (σ a numerical suffix), meaning thereby that q is the letter corresponding to the place σ in the series $1, 2, 3, \dots$. If instead of the indefinite series $(1, b, c, d, \dots)$ we consider, as is sometimes convenient, a definite series of terms $(1, b, c, \dots, q = a_\sigma)$, then σ is said to be the “extent” of the system. The next preceding letter p will naturally be $= a_{\sigma-1}$; and if, increasing the extent by unity, we introduce a new letter r , this will be $a_{\sigma+1}$, and so in other cases, the notation being for the most part used merely as a convenient way of showing the place of a letter in the series.

7. Considering the terms of a given weight, or say a column, in AO or CO , we may represent any portion of the column by means of its initial and final terms, say I and F , by the notations $IaoF$ and $IcoF$ respectively. But a much more important notation is $IcaF$: viz. this represents the series of terms

of given weight which are in *CO* not superior to *I*, and in *AO* not inferior to *F* (a like notation, which however I do not employ, would be *IacF*; viz. this would denote the series of terms which are in *AO* not superior to *I* and in *CO* not inferior to *F*). The definition of *IcaF* has been given in the above general form, but we are in fact exclusively or chiefly concerned with the case where *I* is a non-unitary and *F* a power-ender. It is to be observed that, considering the *AO* column as given, then to form from it the set or interval *IcaF* we may disregard altogether the terms which are in the *AO* column inferior (posterior) to *F*, for by the definition none of these enter into *IcaF*, but it may very well be that there are in *IcaF* terms which are in the *AO* column superior (anterior) to *I*. An instance of this first presents itself for the weight 11; viz. here a portion of the *AO* column is (*fg, b²j, bci, bdh, beg, bf², c²h, cdg, ...*): hence in *IcaF*, if the initial term be *c²h*, for instance in *c²hca b³e²*, we have terms *fg, beg, bf²* which are in *AO* anterior to the initial term *c²h*. In order therefore to form *IcaF* from the *AO* column, we must first take the terms (if any) which being in *CO* posterior to *I* are in the *AO* column anterior to *I*, and then from the portion *IaoF* of the *AO* column reject the terms (if any) which are in *CO* anterior to *I*. In particular, starting from the *AO* column, and

arranging the non-unitaries thereof in *CO* and the power-enders in *AO*, for instance, weight 12, these are

$$\begin{array}{ll} m & g^2 \\ ck & cf^2 \\ eh & b^4e \\ dj & bf^3 \\ fh & c^2e^2 \\ \vdots & \vdots \end{array}$$

There is no difficulty in writing down the terms of the several sets or intervals

$$mca g^2, mca cf^2, mca e^3, \dots, ckca g^2, ckca cf^2, \dots$$

Instead of *ca* we may, if we please, use, and in fact I generally use the conventional symbol ∞ , or write $m \infty g^2, m \infty cf^2$, &c. In any such set, the terms need not be arranged in *AO*; if for any purpose it is more convenient, they may be arranged in *CO*; but of course the definition of the meaning must not be departed from. The expressed initial is the highest term in *CO*, and the expressed final the lowest term in *AO*.

8. I diminish a term by replacing successively each letter thereof by the next inferior letter; for instance, if the term is *cdf*, then the diminished terms are *Dcdf = bdf, c²f, cde*, and so *Db²df = bdf, b²cf, b²de* (where the diminished *b* is *a*, that is, 1). Conversely, we may augment a term by replacing successively each letter thereof by the next superior letter; for instance, *Abdf = b²df, cdf, bef, bdg*, where the first augmentation *b²df* is obtained from the *a* (which may be regarded as latent in the term operated upon). Operating upon the letters in order beginning with the lowest, the several diminutions may be called *D₁, D₂, D₃, ...*, and the several augmentations *A₀, A₁, A₂, ...* (where *A₀* is in fact multiplication by *b*). We diminish a set by diminishing successively the several terms thereof (the diminished terms being taken without repetition; that is, each such term once only). Similarly, we may augment a set by augmenting successively the several terms thereof (the augmented terms being taken without repetition). It is to be noticed that the two operations are not reciprocal to each other; if we diminish a set, and then augment the diminished set, we obtain, indeed all the terms of the original set, but in general we obtain also terms which are not included in the original set.

9. It requires some consideration to see that we have $D(I \infty F) = (D_1 I \infty D_\theta F)$, where $D_\theta F$ is the diminution performed upon the highest letter of *F*. Take any term *M* of $D(I \infty F)$, the several diminutions *D₁M, D₂M, ..., D_θM* are arranged in descending order: *D₁M* the highest and *D_θM* the lowest, as well in *CO* as in *AO*. If then *D₁M* is in *CO* not superior to *D₁I*, then all the *DM*'s will be in *CO* not superior to *D₁I*; and similarly, if *D_θM* is in *AO* not inferior to *D_θF*, then all the *DM*'s will be in *AO* not inferior to *D_θF*. And this being seen, then if we take

N a term of $(D_1 I \infty D_\theta F)$, and consider the successive augmentations *A₀N, A₁N, ..., A_θN* of *N*, then these will be in ascending order *A₀N* the lowest and *A_θN* the highest in *CO* as well as in *AO*. It may happen that *A_θN* or this and neighbouring terms are in *CO* higher than *I*, and that *A₀N* or this and neighbouring terms are in *AO* lower than *F*, but there will always be a term or terms which is or are in *CO* lower than *I* and in *AO* higher than *F*; and thus not only every term of $D(I \infty F)$ will be a term of $(D_1 I \infty D_\theta F)$, but conversely every term of $(D_1 I \infty D_\theta F)$ will be a term of $D(I \infty F)$, and we thus have the required relation $D(I \infty F) = (D_1 I \infty D_\theta F)$.

Symmetric Functions of the Roots. Art. Nos. 10 to 31.

10. We consider a set of roots $\alpha, \beta, \gamma, \delta, \varepsilon, \dots$ either indefinite in number, or else definite, for instance $\alpha, \beta, \gamma, \delta$. The symmetric functions (rational and integral functions) are in the first instance denoted in the usual manner

$$\begin{aligned} S\alpha &= \alpha + \beta + \gamma + \delta + \dots, & S\alpha\beta &= \alpha\beta + \alpha\gamma + \beta\gamma + \dots, \\ S\alpha^2\beta &= \alpha^2\beta + \alpha\beta^2 + \alpha^2\gamma + \alpha\gamma^2 + \beta^2\gamma + \beta\gamma^2 + \dots, \end{aligned}$$

viz. the S refers to all the distinct combinations of like form with the combination ($\alpha, \alpha\beta$, or $\alpha^2\beta$, as the case may be) to which it is prefixed. By omitting the S and instead of the roots considering merely their indices, these same symmetric functions would be 1, 11 (or 1^2), 21, &c., and then if instead of the numbers 1, 2, 3, &c., we introduce the symbolic capital letters $B, C, D, \dots$, the same symmetric functions will be represented as B, B^2, BC , &c. (21, that is, 12 is written as BC , and so in other cases, the letters in alphabetical order). The letters $B, C, D, \dots$ are considered as being of the weights 1, 2, 3, $\dots$ respectively, and thus the symmetric functions of a given degree in the roots are represented by the terms of that weight in the symbolic letters $B, C, D, \dots$; thus the symmetric functions of the degree 4 are E, BD, C^2, B^2C, B^4 ; of course these terms may be arranged in AO or in CO as may be most convenient for the purpose in hand. The capital letters $B, C, D, \dots$ are in fact umbrae, but to avoid confusion with subsequent notations I do not in general thus speak of them. A form such as $S\alpha\beta$ or $S\alpha^2\beta^2$, in which there is no index 1, is said to be *non-unitary*; but a form $S\alpha$ or $S\alpha^2\beta$, in which there is an index = 1 or two or more indices each = 1, is said to be *unitary*: or, what is the same thing, in the symbolic representation by capital letters, the form is non-unitary or unitary according as it does not or does contain the letter B .

11. In the ordinary theory of symmetric functions, we connect the coefficients ($1, b, c, d, \dots$) with the roots ($\alpha, \beta, \gamma, \dots$) by the equation

$$1 + bx + cx^2 + dx^3 + \dots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots,$$

and we thus have

$$\begin{aligned} -b &= \alpha + \beta + \gamma + \dots, = S\alpha, = 1, = B, \\ +c &= \alpha\beta + \alpha\gamma + \beta\gamma + \dots, = S\alpha\beta, = 1^2, = B^2, \\ -d &= \alpha\beta\gamma + \dots, = S\alpha\beta\gamma, = 1^3, = B^3, \\ &\quad \&c., \&c.; \end{aligned}$$

and it is to be remarked that, for any given number of roots, there will be this same number of coefficients: we may for instance have

$$1 + bx + cx^2 + dx^3 = (1 - \alpha x)(1 - \beta x)(1 - \gamma x),$$

that is,

$$\begin{aligned} -b &= \alpha + \beta + \gamma, \\ +c &= \alpha\beta + \alpha\gamma + \beta\gamma, \\ -d &= \alpha\beta\gamma; \end{aligned}$$

and similarly if the number of roots be = 4, or any larger number.

12. The symmetric functions of a given degree, say 4, in the roots, viz.

$$S\alpha^4, S\alpha^3\beta, S\alpha^2\beta^2, S\alpha^2\beta\gamma, S\alpha\beta\gamma\delta,$$

or

$$4, 31, 2^2, 21^2, 1^4,$$

or

$$E, BD, C^2, B^2C, B^4,$$

are equal in number to the combinations of the weight 4 in the coefficients, viz.

$$e, bd, c^2, b^2c, b^4;$$

and the terms of the one set are in fact linear combinations (with mere numerical multipliers) of the terms of the other set; but more than this, we have for instance

$$e = \alpha\beta\gamma\delta + \dots, \text{ that is, } e = B^4;$$

$bd = (\alpha + \beta + \dots)(\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta \dots)$ contains only terms $\alpha^2\beta\gamma$ and $\alpha\beta\gamma\delta$, that is, bd is a linear function of B^2C and B^4 ;

$c^2 = (\alpha\beta + \alpha\gamma + \dots)^2$ contains only terms $\alpha^2\beta^2$, $\alpha^2\beta\gamma$ and $\alpha\beta\gamma\delta$, that is, c^2 is a linear function of C^2 , B^2C and B^4 ; and so on.

13. We have in fact the Table IV (a) which I quote from my paper “A Memoir on the Symmetric Functions of the Roots of an Equation,” *Phil. Trans.*, t. 147 (1857), pp. 489–496, [147],

IV (a).	e	bd	c^2	b^2c	b^4
$S\alpha^4 = 4 = E$					+1
$S\alpha^3\beta = 31 = BD$		+1		+4	+4
$S\alpha^2\beta^2 = 2^2 = C^2$			+1	+2	+6
$S\alpha^2\beta\gamma = 21^2 = B^2C$		+2	+2	+5	+12
$S\alpha\beta\gamma\delta = 1^4 = B^4$	+1	+4	+6	+12	+24

inserting on the left-hand outside margin the new symbols E , BD , &c., with their

explanations: the | indicates that the table is to be read according to the columns, $e = +1B^4$, $bd = +1B^2C + 4B^4$, &c. This table gives conversely a Table IV (b), read according to the lines and serving to express the symmetric functions E , BD , &c., as linear functions of the combinations e , bd , c^2 , b^2c , b^4 of the coefficients.

14. The (a) and (b) tables are given in the Memoir up to X (a) and X (b): it is proper to quote here the (b) tables up to VI (b) with only the change of substituting on the outside left-hand margins the literal terms such as E , BD , &c., instead of the symbols 4, 31, &c., originally used to denote these symmetric functions—it is to be observed that the left-hand symbols are in AO , the upper symbols in CO , this distinction first manifesting itself in the Table VI(b), so that it was necessary to go as far as this in order to put in evidence the true form of the tables.

II (b).				
	c	b^2		
C	−2	+1	III (b).	
B^2	+1			
	d	bc	b^3	
D	−3	+3	−1	IV (b).
BC	+3	−1		
B^3	−1			
	e	bd	c^2	$b^2c\ b^4$
E	−4	+4	+2	−4 +1
BD	+4	−1	−2	+1
C^2	+2	−2	+1	
B^2C	−4	+1		
B^4	+1			

V (b).

	f	be	cd	b^2d	bc^2	b^3c	b^5
F	−5	+5	+5	−5	−5	+5	−1
BE	+5	−1	−5	+3	+1	−1	
CD	+5	−5	−1	+2	+1		
B^2D	−5	+1	+2	−1			
BC^2	−5	+2	+1				
B^3C	+5	−1					
B^5	−1						

VI (b).

	g	bf	ce	d^2	b^2e	bcd	b^3d	c^3	b^2c^2	$b^4c b^6$
G	−6	+6	+6	+3	−6	−12	+6	+2	+9	−6 +1
BF	+6	−1	−6	−3	+3	+7	−3	−2	−2	+1
CE	+6	−6	−1	+2	+2	+3	−3	−2	+1	
D^2	+3	−3	−2	−1	+4	+4	+3	−3		
B^2E	−6	+1	+2	+4	−1	−3	+1			
BCD	−12	+7	+3	+4	−3	0				
B^3D	+6	−3	−3	+3	+1					
C^3	+2	−2	−2	−3						
B^2C^2	+9	−2	+1							
B^4C	−6	+1								
B^6	+1									

It is hardly necessary to remark in relation to these tables that if there are only two roots, then $d = 0$, &c., viz. Table II. is not affected but all the subsequent tables assume a simplified form; if there are only three roots, then $e = 0$, &c., viz. Tables II. and III. are not affected but all the subsequent tables assume a simplified form; and so on.

15. We have between the differential symbols $\partial_b, \partial_c, \partial_d, \dots$ and $\partial_\alpha, \partial_\beta, \partial_\gamma, \dots$ certain relations which it is interesting to develop: it will be convenient to consider successively the cases, three roots, four roots, &c.

In the case of three roots, starting from

$$\begin{aligned} -b &= \alpha + \beta + \gamma, \\ +c &= \alpha\beta + \alpha\gamma + \beta\gamma, \\ -d &= \alpha\beta\gamma, \end{aligned}$$

we have

$$\begin{aligned} \partial_\alpha &= -\partial_b + (\beta + \gamma)\partial_c - \beta\gamma\partial_d, \\ \partial_\beta &= -\partial_b + (\gamma + \alpha)\partial_c - \gamma\alpha\partial_d, \\ \partial_\gamma &= -\partial_b + (\alpha + \beta)\partial_c - \alpha\beta\partial_d, \end{aligned}$$

equations which give conversely $\partial_b, \partial_c, \partial_d$ as linear functions of $\partial_\alpha, \partial_\beta, \partial_\gamma$: I write

down the three equations thus obtained together with a fourth equation which I will explain. The four equations are

$$\begin{aligned} -\partial_b + \delta' &= \frac{\alpha^2}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{\beta^2}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{\gamma^2}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma, \\ -\partial_b &= \frac{\alpha}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{\beta}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{\gamma}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma, \\ -\partial_c &= \frac{\alpha}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{\beta}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{\gamma}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma, \end{aligned}$$

$$-\partial_d = \frac{1}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{1}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{1}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma.$$

In verification of the last three equations, observe that they give

$$-\partial_b + (\beta + \gamma) \partial_c - \beta \gamma \partial_d = \partial_\alpha,$$

that is, $-\partial_b + (\beta + \gamma) \partial_c - \beta \gamma \partial_d = \partial_\alpha$: and similarly from the same three equations we deduce the values of ∂_β and ∂_γ ; the three equations are thus equivalent to the foregoing three equations for $\partial_\alpha, \partial_\beta, \partial_\gamma$.

As to the first equation, to avoid confusion with a root δ , I have written therein δ' (afterwards replaced by δ) to denote the degree of a function homogeneous in (a, b, c, d) , upon which the symbols are supposed to operate; this is also the degree in the roots α, β, γ . The four equations give

$$-a(\partial_\alpha - \partial_\beta + \dots) - b \partial_b - c \partial_c - d \partial_d = \frac{\alpha^3 + b\alpha^2 + c\alpha + d}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \&c., = 0,$$

since

$$\alpha^3 + b\alpha^2 + c\alpha + d = 0, \quad \beta^3 + b\beta^2 + c\beta + d = 0, \quad \gamma^3 + b\gamma^2 + c\gamma + d = 0.$$

The equations thus give

$$\delta \partial_a + b \partial_b + c \partial_c + d \partial_d = \delta,$$

which is right, and the first equation is thus verified.

16. From the last three equations for $\partial_b, \partial_c, \partial_d$, we deduce

$$\begin{aligned} -3 \partial_b - 2\beta \partial_c - \beta^2 \partial_d &= \frac{3\beta^2 + 2b\beta + c}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{3\beta^2 + 2b\beta + c}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{3\gamma^2 + 2b\gamma + c}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma, \\ &= \partial_\alpha + \partial_\beta + \partial_\gamma, \end{aligned}$$

a result more easily deducible from the first set of three equations for $\partial_\alpha, \partial_\beta, \partial_\gamma$: but I have preferred to obtain it in this manner for the sake of the remark that it is a peculiarity of this combination of $\partial_b, \partial_c, \partial_d$ that the coefficients of $\partial_\alpha, \partial_\beta, \partial_\gamma$

become integral functions of the roots (in the actual case constants and $= 1$): for a somewhat similar form

$$-(c \partial_b + b \partial_c) = \frac{\alpha^2 + \alpha\beta + \alpha\gamma}{\alpha - \beta \cdot \alpha - \gamma} \partial_\alpha + \frac{\beta^2 + \beta\gamma + \beta\alpha}{\beta - \gamma \cdot \beta - \alpha} \partial_\beta + \frac{\gamma^2 + \gamma\alpha + \gamma\beta}{\gamma - \alpha \cdot \gamma - \beta} \partial_\gamma,$$

the coefficients are fractional.

We at once have

$$\alpha \partial_\alpha + \beta \partial_\beta + \gamma \partial_\gamma = b \partial_b + 2c \partial_c + 3d \partial_d,$$

viz. these symbols operating upon a function of the roots of the degree ω , or what is the same thing, a function of the coefficients of the weight ω , are each of them equivalent to a constant factor ω .

Again, we have

$$\begin{aligned} \alpha^2 \partial_\alpha + \beta^2 \partial_\beta + \gamma^2 \partial_\gamma &= -(b^2 - 2c) \partial_b - (bc - 3d) \partial_c - bd \partial_d, \\ &= -b(b \partial_b + c \partial_c + d \partial_d) + 2c \partial_b + 3d \partial_c, \end{aligned}$$

or since $a \partial_a + b \partial_b + c \partial_c + d \partial_d = \delta'$ (if as before δ' is the degree of the function operated upon) and therefore $b \partial_b + c \partial_c + d \partial_d = \delta' - a \partial_a$ or say $= \delta' - \partial_a$, this is

$$\alpha^2 \partial_\alpha + \beta^2 \partial_\beta + \gamma^2 \partial_\gamma = -b\delta' + b \partial_a + 2c \partial_b + 3d \partial_c,$$

so that we have here another form $-b\delta' + b \partial_a + 2c \partial_b + 3d \partial_c$, for which the coefficients of $\partial_\alpha, \partial_\beta, \partial_\gamma$ are integral functions of the roots.

17. In the case of four roots, the corresponding equations are

$$\begin{aligned} -b &= \alpha + \beta + \gamma + \delta, \\ +c &= \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta, \\ -d &= \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta, \\ +e &= \alpha\beta\gamma\delta, \end{aligned}$$

and we then have

$$\begin{aligned}\partial_\alpha &= -\partial_b + (\beta + \gamma + \delta)\partial_c - (\beta\gamma + \beta\delta + \gamma\delta)\partial_d + \beta\gamma\delta\partial_e, \\ \partial_\beta &= -\partial_b + (\gamma + \delta + \alpha)\partial_c - (\gamma\delta + \gamma\alpha + \delta\alpha)\partial_d + \gamma\delta\alpha\partial_e, \\ \partial_\gamma &= -\partial_b + (\delta + \alpha + \beta)\partial_c - (\delta\alpha + \delta\beta + \alpha\beta)\partial_d + \delta\alpha\beta\partial_e, \\ \partial_\delta &= -\partial_b + (\alpha + \beta + \gamma)\partial_c - (\alpha\beta + \alpha\gamma + \beta\gamma)\partial_d + \alpha\beta\gamma\partial_e,\end{aligned}$$

and the converse set of equations, which for shortness I write in the form

$$\begin{aligned}-\partial_b + \delta' &= -\partial_b, -\partial_c, -\partial_d, -\partial_e \\ &= \frac{\alpha^{4,3,2,1,0}}{\alpha - \beta \cdot \alpha - \gamma \cdot \alpha - \delta} \partial_\alpha + \frac{\beta^{4,3,2,1,0}}{\beta - \gamma \cdot \beta - \delta \cdot \beta - \alpha} \partial_\beta + \frac{\gamma^{4,3,2,1,0}}{\gamma - \delta \cdot \gamma - \alpha \cdot \gamma - \beta} \partial_\gamma + \frac{\delta^{4,3,2,1,0}}{\delta - \alpha \cdot \delta - \beta \cdot \delta - \gamma} \partial_\delta.\end{aligned}$$

We have, in like manner as in the former case,

$$\begin{aligned}-4\partial_b - 3\partial_c - 2c\partial_d - d\partial_e &= \partial_\alpha + \partial_\beta + \partial_\gamma + \partial_\delta, \\ b\partial_b + c\partial_c + d\partial_d + e\partial_e &= \alpha\partial_\alpha + \beta\partial_\beta + \gamma\partial_\gamma + \delta\partial_\delta = \omega, \\ -b\delta' + b\partial_a + 2c\partial_b + 3d\partial_c + 4e\partial_d &= \alpha^2\partial_\alpha + \beta^2\partial_\beta + \gamma^2\partial_\gamma + \delta^2\partial_\delta;\end{aligned}$$

and similarly in the case of five or more roots.

18. In the case of σ roots, I write $m = a_\sigma$, and for shortness

$$\begin{aligned}\Theta_\sigma &= c\partial_b + (\sigma' - 1)d\partial_c + \dots + a'm\partial_l, \\ P &= b\partial_a + 2c\partial_b + 3d\partial_c + \dots + \sigma'm\partial_l,\end{aligned}$$

so that, besides the equation $b\partial_b + c\partial_c + \dots + m\partial_m = S\alpha\partial_\alpha = \omega$, the foregoing investigations show that we have

$$\begin{aligned}\Theta_\sigma &= -S\partial_\alpha, \\ P - b\delta &= S\alpha^2\partial_\alpha.\end{aligned}$$

The operand for these symbols is a symmetric function of the roots, which is thus also a function of the coefficients: it is of the degree ω in the roots, and consequently of the weight ω in the coefficients, and its degree in the coefficients is taken to be $= \delta$. It is sometimes convenient to represent this operand, *qua* function of the roots, by T and, *qua* function of the coefficients, by U , so that we have in general $T = U$. If T be a non-unitary function of the roots, then we may say that $T, = U$, is a non-unitariant.

19. I give some illustrations of the equation $\Theta_\sigma = -S\partial_\alpha$. Suppose

$$T = U = S\alpha^4 = E = -4e + 4bd + 2c^2 - 4b^2c + b^4$$

(Table IV (b)); σ' must be $= 4$ at least and I take it to be 4 and 5 successively; we thus have

$$\Theta_4 = 4\partial_b + 3c\partial_c + 2c\partial_d + d\partial_e,$$

$$\Theta_5 = 5\partial_b + 4bd\partial_c + 3c\partial_d + 2d\partial_e,$$

omitting from Θ_5 the term $e\partial_f$ which is obviously inoperative. For any number whatever of roots, we have

$$-S\partial_\alpha \cdot S\alpha^4 = -4S\alpha^3 = -4(-d + 3bc - b^3) = 12d - 12bc + 4b^3,$$

and this should therefore be the value as well of Θ_4E as of Θ_5E . The calculations may be arranged as follows:

$$\begin{array}{rcl}\Theta_4E: & & \\ 4 \cdot 4d - 8bc + 4b^3 & d + 16 & -4 \quad 12 \\ 3b \cdot 4c - 4b^2 & bc - 32 + 12 + 8 & -12 \quad + 4 \\ 2c \cdot 4b & b^3 + 16 - 12 & \\ d \cdot -4 & & \end{array}$$

$$\begin{array}{l} \Theta_5 E: \\ 5 \cdot 4d - 8bc + 4b^3 \qquad \qquad \qquad d - 20 \qquad -8 \quad 12 \\ 4b \cdot 4c - 4b^2 \qquad \qquad bc + 40 - 16 - 12 \quad -12 \quad +4 \\ 3c \cdot 4b \qquad \qquad \qquad b^3 + 20 - 16 \\ d \cdot -4 \end{array}$$

giving in each case the right result.

20. In the foregoing example, $S\alpha^4$ was a non-unitary function of the roots, but I take the case of a unitary function. Suppose

$$T = U = S\alpha^3\beta = BD = 4e - bd - 2c^2 + b^2c.$$

Here $-S\partial_\alpha \cdot S\alpha^3\beta$ is not independent of the number of the roots; in the case of 4 roots, we have

$$-S\partial_\alpha \cdot S\alpha^3\beta = -3S\alpha^2\beta - 3S\alpha^3 = -3(3d - bc) - 3(-3d + 3bc - b^3) = 0 \cdot d - 6bc + 3b^3;$$

and in the case of 5 roots, we have

$$-S\partial_\alpha \cdot S\alpha^3\beta = -3S\alpha^2\beta - 4S\alpha^3 = -3(3d - bc) - 4(-3d + 3bc - b^3) = 3d - 9bc + 4b^3;$$

and these should therefore be the values of $\Theta_4 BD$ and $\Theta_5 BD$ respectively. The calculations are

$$\begin{array}{l} \Theta_4 BD: \\ 4 \cdot -d + 2bc \qquad \qquad \qquad d - 4 \qquad -6 \quad 0 \\ 3b \cdot -4c + b^2 \quad bc + 8 - 12 - 2 \qquad \qquad +3 \\ 2c \cdot -b \qquad \qquad \qquad b^3 + 3 \\ d \cdot +4 \end{array}$$

$$\begin{array}{l} \Theta_5 BD: \\ 5 \cdot -d + 2bc \qquad \qquad \qquad d - 5 \quad +8 \quad +3 \\ 4b \cdot -4c + b^2 \quad bc + 10 - 16 - 3 \quad -9 \quad -3 \\ 3c \cdot -b \qquad \qquad \qquad b^3 + 4 \qquad \qquad +4 \\ 2d \cdot +b \end{array}$$

giving in each case the correct result. We have $\Theta_5 - \Theta_4 = \partial_b + b\partial_c + c\partial_d + d\partial_e$, and the examples show that performing this operation on the non-unitariant $S\alpha^4$, $= E$, we obtain a result $= 0$; whereas for the unitary function $S\alpha^3\beta$, $= BD$, the result is not $= 0$.

21. Considering the question generally, I take the highest coefficient in U to be $q = a_\sigma$ (σ equal to or less than ω), or what is the same thing, the extent of U to be $= \sigma$; this implies that σ' is at least $= \sigma$; and taking it to be first $= \sigma$, and then to be any number greater than σ , we have

$$\Theta_\sigma = -S\partial_\alpha, \quad \Theta_{\sigma'} = -S\partial_\alpha.$$